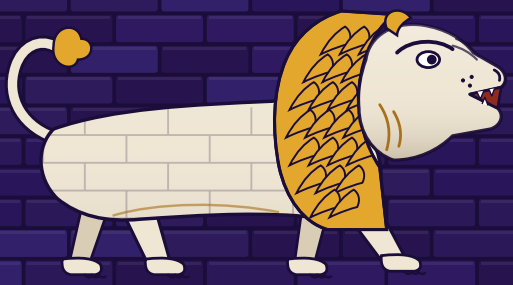
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

EN

HUMANS AND THE ENVIRONMENT

A handout of ideas, facts and problems



8 sections · Edition of November 2026 · 26 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Humans and the Environment

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

1 Environment and ecosystem components

Aims. Classify living and nonliving ecosystem components; Explain one interaction linking an organism with an abiotic factor.
Before this section: An ecosystem includes organisms and the physical conditions around them.

An ecosystem combines organisms with the physical and chemical surroundings with which they interact. Biotic factors are living components; abiotic factors include light, temperature, water and minerals. A population is one species in an area; a community includes interacting populations.

Idea 1. An ecosystem includes a living community and the conditions through which it interacts. Sunlight and minerals are not alive, but they affect what organisms can do. An abiotic factor is therefore not an unimportant factor; it is simply a nonliving part of the system.

Fact 1.

$$\text{Organism} \rightarrow \text{population} \rightarrow \text{community} \rightarrow \text{ecosystem}$$

Why it holds. Living status determines the basic biotic/abiotic classification. Nonliving conditions can still strongly affect organism growth, survival and interactions.

Method 1.

1. Name the ecosystem and define the area being considered.
2. Sort its organisms as biotic and its physical or chemical conditions as abiotic.
3. Choose one organism and explain how a named abiotic factor affects it.
4. Use a causal sentence rather than an unconnected list of examples.

Recognising it. For classification, ask whether the item is an organism or a physical/chemical environmental factor.

Pr 1. Fish, algae and bacteria are biotic components of a pond.

Solution. Fish, algae and bacteria are organisms, so they are biotic pond components. Water temperature and dissolved oxygen are abiotic conditions even though living things can affect them. Fish depend on sufficient dissolved oxygen for respiration. The example therefore shows both a classification and an interaction between the living community and its physical environment.

Common error. A habitat is where an organism lives; an ecosystem includes the interacting system.

In the examination. For explain questions add an interaction after classifying: name the factor, the organism and the effect.

Pr 2. Are bacteria abiotic because they are microscopic?

Pr 3. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

Pr 4. Soil mineral content affects plant growth. Why is it still abiotic?

(a) It is a nonliving chemical condition

(b) Anything affecting plants is biotic

(c) All material found in soil is living

2 Energy flow, food chains and decomposers

Aims. Assign producer and consumer roles within a stated food chain; Distinguish energy flow from nutrient cycling.
Before this section: Producers capture energy; consumers obtain energy by feeding; decomposers break down organic material.

Producers use light or chemical energy to make organic material from inorganic substances. Consumers obtain organic material by feeding. Decomposers break down dead material and help recycle nutrients. Energy passes through feeding levels and much becomes dispersed heat. Nutrients cycle, but usable energy is not recycled in the same way.

Idea 2. Food-chain energy is a flow through successive roles. Only some energy at one trophic level becomes available to the next, because organisms use energy and not all material is eaten or assimilated. A specified transfer percentage is a model instruction for the calculation, not a universal ecological constant.

Fact 2.

$$\text{Producer} \rightarrow \text{primary consumer} \rightarrow \text{secondary consumer}$$

Why it holds. A percentage must be applied to the energy available at the immediately preceding level. Repeating a fractional transfer reduces energy successively rather than subtracting the same original amount each time.

Method 2.

1. Start the chain with its producer and draw arrows in the direction of food-energy transfer.
2. Assign each consumer level from what that organism eats in the stated chain.
3. Explain that energy is used and dissipated at each transfer.

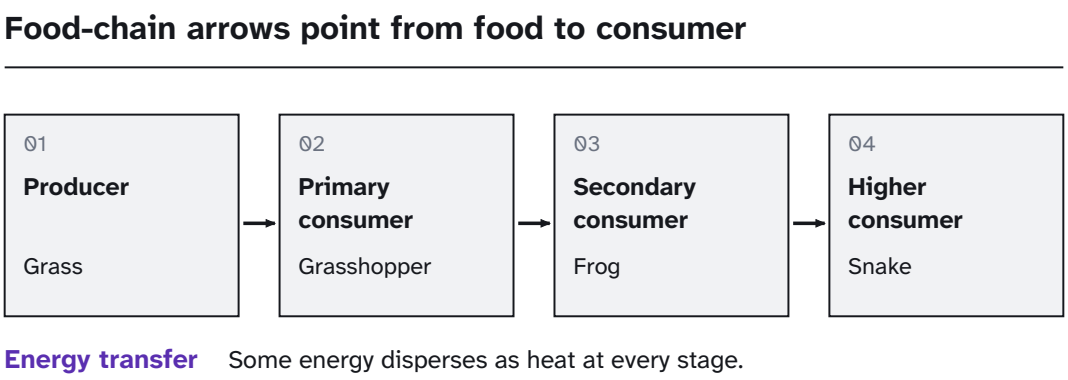


Figure 1. Figure 07. The arrow direction is food to consumer: grass to grasshopper to frog to snake. This is one simplified chain within a food web. Usable energy decreases along the chain; nutrients can be recycled.

4. Describe decomposers returning nutrients to the environment without claiming energy is recycled.

Recognising it. If a question explicitly gives transfer efficiency, apply it once for every trophic-level step.

Pr 5. In grass to grasshopper to frog, grass is the producer, grasshopper the primary consumer and frog the secondary consumer.

Solution. Grass captures light energy and makes organic material, so it is the producer. The grasshopper eats the producer and is the primary consumer. The frog eats that consumer and is the secondary consumer. The arrows grass → grasshopper → frog show transfer toward the eater; dead material from all levels can enter decomposer pathways.

Common error. The 10 percent rule is a rough teaching approximation, not a universal exact constant.

In the examination. Consumer levels are roles in the specified chain; do not assign a permanent level without looking at the feeding relationship.

Pr 6. Does energy cycle back through the ecosystem in the same way as mineral nutrients?

Pr 7. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

Pr 8. A question assumes 10 percent transfer at every step. What fraction of producer energy reaches secondary consumers?

(a) 20 percent

(b) 1 percent

(c) 10 percent

3 Population growth and limiting factors

Aims. Calculate net population change from four demographic processes; Explain how a limiting factor can constrain population growth.

Before this section: Population means members of one species in a defined area and period; basic signed addition.

Population size changes through births, deaths, immigration and emigration. Rapid exponential growth can occur when resources are abundant. Limited food, space and other conditions constrain growth. Carrying capacity describes a population size an environment can sustain under specified conditions.

Idea 3. A population boundary makes gains and losses countable. Births and immigration add members inside the boundary; deaths and emigration remove them. Net change is the balance of those processes, while final population also includes the members already present at the start.

Fact 3.

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

Why it holds. Immigration and births have the same positive effect on local count despite different origins. A positive net change must be added to the initial population rather than reported as the final size.

Method 3.

1. Set the population boundary and time interval.
2. Treat births and immigration as additions and deaths and emigration as removals.
3. Add the gains and subtract the losses to obtain net change.
4. If explaining growth limits, link a specific resource or condition to survival or reproduction.

Recognising it. A starting population plus four demographic counts calls for a change calculation followed by a final-size calculation.

Pr 9. If births are 30, deaths 10, immigration 5 and emigration 8, the population increases by 17.

Solution. The example gives 30 births and 5 immigrants, so total additions are 35. It gives 10 deaths and 8 emigrants, so total removals are 18. Net change is $35 - 18 = +17$ individuals. This is the increase during the interval, not the final population; a starting population would be needed to calculate the latter.

Common error. Carrying capacity is not an unchanging property independent of habitat conditions.

In the examination. Label a positive result increase and a negative result decrease; do not report net change as the final population size.

Pr 10. If births equal deaths, must population size stay unchanged?

Pr 11. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

Pr 12. Births equal deaths, but immigration exceeds emigration. The local population:

(a) Stays constant

(b) Must decrease

(c) Increases

4 Relationships among organisms

Aims. Identify ecological interactions from their effects on each partner; Support an interaction label with scenario evidence.
Before this section: Benefit, harm and little measurable effect are relative to the organisms in the interaction.

Competition occurs when organisms use a limited shared resource. Predation involves one organism consuming another. In mutualism both benefit; in commensalism one benefits without a substantial effect on the other; in parasitism one benefits at the host expense.

Idea 4. An interaction label summarizes what happens to each partner compared with the relevant situation without the interaction. A benefit for one partner does not automatically mean a benefit for the other. Writing the two effects separately is more reliable than guessing a label from the organisms involved.

Fact 4.

Mutualism $(+,+)$
commensalism $(+,0)$
parasitism $(+,-)$

Why it holds. Mutualism requires a benefit to both partners. A benefit/harm pattern differs from a benefit/no-clear-effect pattern, so the second partner cannot be ignored.

Method 4.

1. Identify the two organisms and the resource or activity linking them.
2. Describe the effect on the first organism and on the second separately.
3. Match the two effects to the appropriate interaction category.
4. Use evidence from the case rather than assuming all close associations are mutualistic.

Recognising it. For relationship scenarios, identify the sign of the effect on each organism before naming the interaction.

Pr 13. A parasite obtaining nutrients while harming its host is a plus-minus interaction.

Solution. The parasite obtains nutrients, so it benefits from the interaction. The host is harmed, giving the benefit/harm pattern commonly written plus/minus. The two competing plants both face restricted access to limited light. Competition is therefore a minus/minus interaction in the comparison with unrestricted resource access, even if one competitor does better than the other.

Common error. A relationship may be context-dependent; answer using the effects given in the question.

In the examination. Write the effect on each partner before naming the interaction; this makes similar scenarios easier to distinguish.

Pr 14. Does one competitor winning mean competition is automatically a benefit/harm interaction?

Pr 15. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Pr 16. One organism benefits while the other has no clear benefit or harm. Is this necessarily parasitism?

(a) No; parasitism requires harm to the host

(b) Yes; any one-sided benefit is parasitism

(c) Yes; no effect means harm

5 Pollution: sources, pathways and effects

Aims. Build a source-pathway-receptor explanation of pollution harm; Select prevention that interrupts a stated nutrient-runoff pathway.
Before this section: Nutrients support growth; decomposition can consume dissolved oxygen.

Pollution changes environmental conditions in ways that can cause harm or reduce usefulness. Sources can be natural or human-made. Air, water and soil pollution are connected through transport and deposition. A useful answer links a source to a pathway, an effect and a control.

Idea 5. Pollution explanations are strongest when each link shows how one change causes the next. Fertilizer does not have to poison fish directly to harm them: nutrients can change biological growth, and subsequent decomposition can alter oxygen conditions. Prevention can target an earlier link before ecological damage occurs.

Fact 5.

$$\text{Source} \rightarrow \text{pathway} \rightarrow \text{receptor} \rightarrow \text{effect}$$

Why it holds. Runoff transports nutrients from a land source into water. Decomposition of increased organic material can use dissolved oxygen and harm oxygen-dependent

organisms.

Method 5.

1. Name the pollution source and the released material.
2. Trace how the material reaches the affected ecosystem.
3. Explain the intermediate process connecting exposure to the final effect.
4. Choose a preventive measure and state which link it reduces.

Recognising it. A source-to-effect question needs an intermediate pathway and mechanism, plus prevention linked to that chain.

Pr 17. Nutrient-rich runoff can stimulate excessive algal growth.

Solution. Excess nutrients from fertilizer can be carried by runoff into a water body. Nutrient enrichment can stimulate excessive algal growth. When organic material is decomposed, microbial respiration can consume dissolved oxygen. Low oxygen can harm fish; reducing excess fertilizer or intercepting runoff addresses the source or pathway before that effect occurs.

Common error. Do not say algae always produce an immediate oxygen shortage without explaining decomposition and conditions.

In the examination. Use connecting verbs such as carries, stimulates and consumes to make every step in the causal chain explicit.

Pr 18. Why is algae cause fish death an incomplete explanation?

Pr 19. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

Pr 20. Which measure most directly interrupts the pathway from fertilized land to a nearby stream?

(a) Counting fish without changing the source or pathway

(b) Intercepting nutrient-rich runoff before it reaches the stream

(c) Adding more fertilizer whenever algae increase

6 Ecological footprint and biocapacity

Aims. Compare ecological demand with biocapacity on a common basis; Interpret a numerical balance as a deficit or reserve.

Before this section: Demand versus available capacity; subtraction; per-person and total quantities must not be mixed.

Ecological footprint estimates demand for biologically productive area needed to support consumption and absorb certain wastes. Biocapacity represents available biological production capacity. A deficit occurs when demand exceeds capacity. A global hectare is a productivity-adjusted unit, not simply any local hectare.

Idea 6. Footprint and biocapacity compare demand with biological supply within an accounting model. A deficit means the demand figure exceeds the available capacity figure on the same basis. The calculation only makes sense if both quantities describe the same population basis and productivity-adjusted unit.

Fact 6.

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$
$$\text{if positive}$$

Why it holds. Subtracting capacity from demand tests whether demand exceeds capacity. A positive gap indicates deficit; the reverse comparison would indicate a reserve rather than a negative physical area.

Method 6.

1. Identify footprint as demand and biocapacity as available biological production capacity.
2. Check that both figures use the same units, time basis and population basis.
3. Subtract biocapacity from footprint to test for a deficit.
4. Explain the result within the indicator scope rather than treating it as every environmental impact.

Recognising it. For a footprint balance, check units and per-person versus total basis before subtracting and interpreting.

Pr 21. A population with footprint 3 global hectares per person and biocapacity 2 has a deficit of 1 per person.

Solution. The example footprint is 3 global hectares per person, describing demand. Biocapacity is 2 global hectares per person, describing available capacity on the same basis. Demand exceeds capacity by $3 - 2 = 1$ global hectare per person. The result is an ecological deficit, not an area measured in ordinary unadjusted local hectares.

Common error. Ecological footprint and carbon footprint are related but not identical measures.

In the examination. Write global hectares per person in a per-person result; omitting the population basis changes its meaning.

Pr 22. Can a region total footprint be compared directly with biocapacity per person?

Pr 23. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

Pr 24. A region has footprint lower than its biocapacity on the same basis. This indicates:

(a) An ecological deficit

(b) Proof that every environmental impact is zero

(c) An ecological reserve in this accounting comparison

7 Waste management and the hierarchy

Aims. Distinguish prevention, reuse, recycling, recovery and disposal; Apply the waste hierarchy to a concrete material scenario.
Before this section: Reuse keeps an item in service; recycling reprocesses its material.

Waste management involves preventing, collecting, sorting, treating and disposing of waste. Prevention and reduction usually come before reuse, recycling, recovery and final disposal. Hazardous items require appropriate separate handling; not every household discard belongs in an ordinary bin.

Idea 7. The waste hierarchy separates avoiding a discard from keeping an item in use and from reprocessing its material. These actions can all reduce disposal, but they act at different stages. Correct classification follows what actually happens to the item, not a general impression that the action is environmentally helpful.

Fact 7.

$$\text{Prevent/reduce} \rightarrow \text{reuse} \rightarrow \text{recycle} \rightarrow \text{recover} \rightarrow \text{dispose}$$

Why it holds. Prevention avoids waste generation before later treatment is needed. Reuse keeps the item in service, whereas recycling reprocesses its material into a new product.

Method 7.

1. Identify whether the action avoids waste, keeps the item in use or processes discarded material.
2. Classify the action using the general hierarchy.
3. For a proposed choice consider prevention or reduction first where feasible.
4. Check material hazards and the approved local collection stream before actual disposal.

Recognising it. Inspect the action verb in a scenario: avoid, use again, reprocess or dispose.

Pr 25. Refilling a reusable bottle is reuse.

Solution. Refilling a durable bottle keeps the same item in service, so it is reuse. Processing separated used glass into new glass changes the material into a new product, so it is recycling. Avoiding unnecessary packaging prevents that material becoming waste at all. These actions belong at different stages of the hierarchy and should not all be labelled recycling.

Common error. Recycling does not erase all upstream resource use or justify unlimited consumption.

In the examination. Use the action in the scenario as evidence: avoid means prevention, use again means reuse, reprocess means recycling.

Pr 26. Is using a clean jar again to store suitable items recycling?

Pr 27. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

Pr 28. A metal drink container is melted and made into a new metal product. This is:

(a) Recycling

(b) Reuse of the same item

(c) Waste prevention

8 Sustainable development and integrated decisions

Aims. Assess a project across environmental, social and economic dimensions; State measurable indicators and a trade-off for a sustainability decision.

Before this section: Present and future needs; a benefit in one dimension can have costs in another.

Sustainable development addresses present needs while preserving future ability to meet needs. Environmental, social and economic dimensions interact. A project can improve one dimension while creating costs in another, so evaluation should consider benefits, trade-offs and long-term effects.

Idea 8. Sustainability decisions combine outcomes that do not all move together. An energy-efficiency project may save electricity and money while requiring upfront spending or causing temporary disruption. Indicators make these claims testable; a trade-off explains how a benefit in one area comes with a cost or constraint elsewhere.

Fact 8.

$$\text{Environment} + \text{society} + \text{economy}$$

Why it holds. Environmental, social and economic effects need distinct evidence. Comparing effects over time prevents an upfront cost or one immediate benefit from dominating the whole judgement.

Method 8.

1. Define the project and the needs it is intended to meet.
2. Give one environmental, one social and one economic effect.
3. Select an indicator that could test each effect.
4. Explain one trade-off over the project life rather than declaring the project sustainable from a single benefit.

Recognising it. When evaluating a project, choose one measurable indicator per dimension and connect a concrete benefit with a concrete cost.

Pr 29. A public-transport project can reduce emissions, improve access and lower travel costs, but its construction and operation also require resources and funding.

Solution. The public-transport example can reduce transport emissions if its use changes travel patterns. Improved access to education or work is a social benefit to evaluate. Travel costs, operating costs and funding provide economic questions. Construction also consumes resources and money, so a full assessment compares these costs with benefits over time.

Common error. Do not equate sustainable development with only economic growth or only nature protection.

In the examination. Replace vague words such as greener with a measurable indicator such as electricity use, emissions, access or cost over a stated period.

Pr 30. Is lower operating energy alone enough to prove a project is sustainable?

Pr 31. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Pr 32. Which is a measurable social indicator for a campus energy-efficiency project?

(a) Electricity use in kilowatt-hours alone

(b) Student-reported thermal comfort using a defined survey

(c) The project is very green

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Use living organism versus nonliving environmental factor as the test, not whether the item is useful. (2) Consider sunlight and mineral content separately from grass and rabbit. (3) Assign each of the four items a category and give a short reason for any item you initially found ambiguous.

Pr 7 (p. 4) (1) Identify the two steps: producers to primary consumers, then primary to secondary. (2) For the first transfer apply the stated 10 percent to 10000 energy units. (3) For the second transfer apply 10 percent to the first result, not to the original producer amount. Label the two consumer-level energies.

Pr 11 (p. 6) (1) Separate the starting 200 individuals from changes during the period. (2) Add births and immigrants, then subtract deaths and emigrants to obtain net change. (3) Add that net change to 200. Report final size as individuals rather than confusing it with the smaller change value.

Pr 15 (p. 7) (1) Choose one relationship with a benefit for each partner and one with benefit for one and harm for the other. (2) Name the appropriate relationship category for each chosen example. (3) Explain the two signs in words, showing what each partner gains or loses rather than giving an unexplained plus/minus label.

Pr 19 (p. 8) (1) Begin with excess fertilizer and explain how its nutrients reach water. (2) Link nutrient enrichment to excess growth and then decomposition-related oxygen depletion under suitable conditions. (3) Finish with fish harm and one measure that reduces the source or intercepts the pathway. Use connected causal statements.

Pr 23 (p. 10) (1) Identify which of 4.5 and 2.8 describes demand and which describes available capacity. (2) Compare them to decide deficit or reserve before calculating the gap. (3) Subtract the smaller from the larger for its magnitude and retain global hectares per person. State the type of balance explicitly.

Pr 27 (p. 11) (1) Read each action separately rather than giving one category for the whole list. (2) Ask whether the action avoids material use, keeps the item in service, reprocesses material or places a residue in final disposal. (3) Match those distinctions to packaging, jar, cans and landfill in order. Use a short label and explanatory action for each.

Pr 31 (p. 12) (1) List the three dimensions and identify a campus outcome for each. (2) Choose measures such as electricity consumption, occupant comfort and financial savings or payback, giving a clear basis for comparison. (3) Add a trade-off, for example upfront spending versus later savings, and explain what decision needs evaluation rather than merely listing two unrelated facts.

Answers

Pr 2 (p. 3) No. Bacteria are living organisms and therefore biotic components.

Pr 3 (p. 3) Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.

Pr 4 (p. 3) (a) Correct. Influence on life does not itself make a factor alive.

Pr 6 (p. 4) No. Energy flows through and is eventually dissipated as heat; nutrients can be recycled.

Pr 7 (p. 4) Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.

Pr 8 (p. 4) (b) Correct. Two transfers multiply one tenth by one tenth.

Pr 10 (p. 6) No. Immigration and emigration can still produce a net change.

Pr 11 (p. 6) Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.

Pr 12 (p. 6) (c) Correct. The birth/death balance is zero, leaving a positive migration balance.

Pr 14 (p. 7) No. Both incur the effects of competing for a limited resource; one may still outperform the other.

Pr 15 (p. 7) Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Pr 16 (p. 7) (a) Correct. Benefit with little or no effect describes a different pattern from benefit with harm.

Pr 18 (p. 8) It omits the conditions and intermediate processes, especially decomposition and dissolved-oxygen depletion.

Pr 19 (p. 8) Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.

Pr 20 (p. 8) (b) Correct. This acts on transport between source and receptor.

Pr 22 (p. 10) No. Convert both to totals or both to per-person values first.

Pr 23 (p. 10) It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.

Pr 24 (p. 10) (c) Correct. Capacity exceeds the counted demand.

Pr 26 (p. 11) It is reuse, because the same item is used again without material reprocessing.

Pr 27 (p. 11) Prevention, reuse, recycling and disposal, respectively. One mark for each classification.

Pr 28 (p. 11) (a) Correct. The discarded material is reprocessed into a new product.

Pr 30 (p. 12) No. Social access, economic viability, construction impacts and long-term effects also require assessment.

Pr 31 (p. 12) Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Pr 32 (p. 12) (b) Correct. A defined comfort measure directly examines the learner experience.

Humans and the Environment, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is an abiotic factor?

   (a) Algae
   (b) Bacteria
   (c) Water temperature
   (d) Fish

2. Decomposers mainly help by:

   (a) Recycling nutrients from dead matter
   (b) Creating sunlight
   (c) Stopping all energy loss
   (d) Removing all producers

3. Which directly reduces a local population?

   (a) Reproduction
   (b) Births
   (c) Immigration
   (d) Emigration

4. Both organisms benefit in:

   (a) Parasitism
   (b) Mutualism
   (c) Predation
   (d) Competition

Part II. Show your reasoning.

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Excess nutrient input into water is most closely associated with:

   (a) Eutrophication
   (b) Lens refraction
   (c) Sterilization
   (d) No biological change

2. A footprint larger than biocapacity indicates:

   (a) Unlimited sustainability
   (b) An ecological reserve
   (c) An ecological deficit
   (d) Zero resource use

3. Which is highest in the general waste hierarchy?

   (a) Disposal after mixing
   (b) Landfill
   (c) Uncontrolled burning
   (d) Waste prevention

4. A complete sustainability assessment considers:

   (a) Only this month
   (b) Environmental, social and economic effects
   (c) Only profit
   (d) Only tree numbers

Part II. Show your reasoning.

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Humans and the Environment, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) Temperature is a physical condition rather than a living organism.

Written part. Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each. (Section: p. 2.)

Question 2. (a) They break down organic matter and return nutrients to the system.

Written part. Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency. (Section: p. 3.)

Question 3. (d) Emigration removes individuals from the local population.

Written part. Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total. (Section: p. 5.)

Question 4. (b) Mutualism is defined by benefits to both participants.

Written part. Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation. (Section: p. 6.)

Humans and the Environment, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.

Written part. Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention. (Section: p. 7.)

Question 2. (c) Demand exceeds the available biological capacity under the accounting model.

Written part. It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation. (Section: p. 9.)

Question 3. (d) Avoiding waste reduces the need for later treatment and disposal.

Written part. Prevention, reuse, recycling and disposal, respectively. One mark for each classification. (Section: p. 10.)

Question 4. (b) The concept integrates all three dimensions and future consequences.

Written part. Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item. (Section: p. 11.)

Facts at a glance

1. Environment and ecosystem components p. 2

$$\text{Organism} \rightarrow \text{population} \rightarrow \text{community} \rightarrow \text{ecosystem}$$

2. Energy flow, food chains and decomposers p. 3

$$\text{Producer} \rightarrow \text{primary consumer} \rightarrow \text{secondary consumer}$$

3. Population growth and limiting factors p. 5

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

4. Relationships among organisms p. 6

$$\text{Mutualism } (+,+)$$
$$\text{commensalism } (+,0)$$
$$\text{parasitism } (+,-)$$

5. Pollution: sources, pathways and effects p. 7

$$\text{Source} \rightarrow \text{pathway} \rightarrow \text{receptor} \rightarrow \text{effect}$$

6. Ecological footprint and biocapacity p. 9

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$
$$\text{if positive}$$

7. Waste management and the hierarchy p. 10

$$\text{Prevent/reduce} \rightarrow \text{reuse} \rightarrow \text{recycle} \rightarrow \text{recover} \rightarrow \text{dispose}$$

8. Sustainable development and integrated decisions p. 11

$$\text{Environment} + \text{society} + \text{economy}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Index

Topics and quantities, with the page where each is introduced. Quantities show their symbol; the Index of notation lists symbols in alphabetical order.

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Index of notation

Latin letters first, then Greek letters, then differences (Δ). The same letter can name different quantities in different sections.